Smart Gossyculture: AI-Powered Cotton Harvesting and Pest Detection System

Problem Statement

Cotton farmers in India, especially those with small land holdings, face many difficulties during the harvesting season. Manual cotton picking is time-consuming, expensive, and often leads to damage to cotton quality. Due to labour shortages, many farmers are forced to delay harvesting, which reduces their yield. Pest attacks and crop diseases are common, but identifying and treating them early is not easy without expert help. Overuse of pesticides is another issue, as farmers tend to spray chemicals on the entire field even if only a small area is affected. Most farmers also don't have access to real-time information about their crops or any digital support system that speaks their language. These challenges result in reduced productivity, poor-quality cotton, and low income.

Innovative Solution

We are building **Smart Gossyculture**, a smart cotton harvesting system that uses a combination of AI, robotics, sensors, and solar power.

The system uses a small camera and an AI model to identify which cotton bolls are mature and ready to be picked. Before harvesting, a moisture sensor checks if the cotton is dry enough. A soft robotic arm carefully plucks the cotton without damaging it. If pests or diseases are found through image analysis, a micro-sprayer automatically applies pesticide only to that area, avoiding unnecessary chemical usage.

Importantly, this system is not meant to replace farmers or reduce their earnings. Instead, it is a tool that assists farmers, helps them to reduce their workload, improve cotton quality, and avoid losses caused by delay, pest damage, or over-spraying. It enables them to make better decisions with real-time data and spend more time on other essential farm work or income sources.

The machine provides voice alerts in the farmer's local language, making it friendly for rural users. All key information — boll count, moisture levels, pest alerts, and spraying records — is sent to a simple mobile app dashboard.

Highlighting, buyers and cotton industries can access harvest stock updates, helping build a transparent farm-to-factory chain through the app.

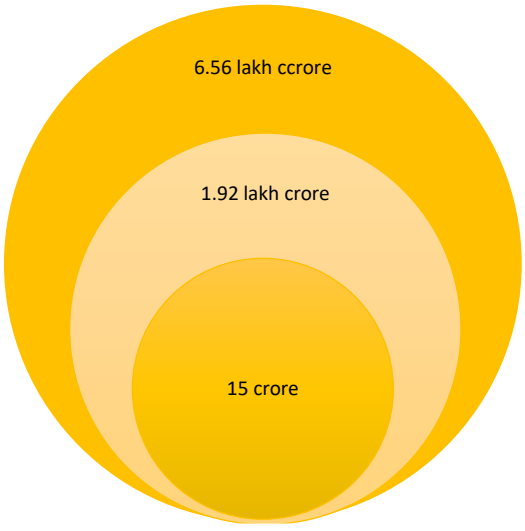
In the future, we are planning to enhance this robot by adding GPS-based navigation for fully autonomous movement, multi-row cotton picking, and integration with farm management apps. We also aim to expand the AI model to detect more diseases and pest types for a wider range of crops.

Market Potential

India has the largest cotton-growing area in the world, with millions of farmers, many of them relying on seasonal labour and traditional methods. The goal of Smart Gossyculture is not to take away jobs, but to make cotton farming easier, faster, and more profitable.

In many areas, labour shortages are already making it hard to harvest on time. Our system gives farmers the option to automate only the most labour heavy tasks, without replacing the need for human involvement. It also creates opportunities for local technicians and youth to operate and maintain the system.

With its affordable and easy-to-use design, this solution can help lakhs of smallholder farmers, while still keeping them at the centre of the process.



Total Addressable Market (TAM):

- **Total cotton farming area globally**: ~33 million hectares (~82 million acres)
- **Interested in this solution**: **80%**
- **TAM = 82 million acres × 80% = 65.6 million acres**
- **Estimated value per acre** (AI harvesting, pest detection, dashboard): ₹10,000

TAM = 82 million acres × 80% = 65.6 million acres
TAM ≈ ₹6.56 lakh crore

Serviceable Available Market (SAM) – India:

- **Cotton farming area in India**: ~13 million hectares (~32 million acres)
- **Interested in this solution**: **60%**

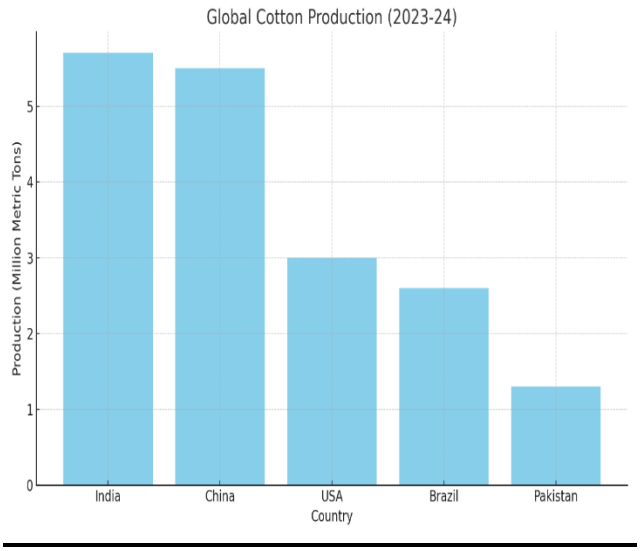
SAM = 32 million acres × 60% = 19.2 million acres
SAM ≈ ₹1.92 lakh crore

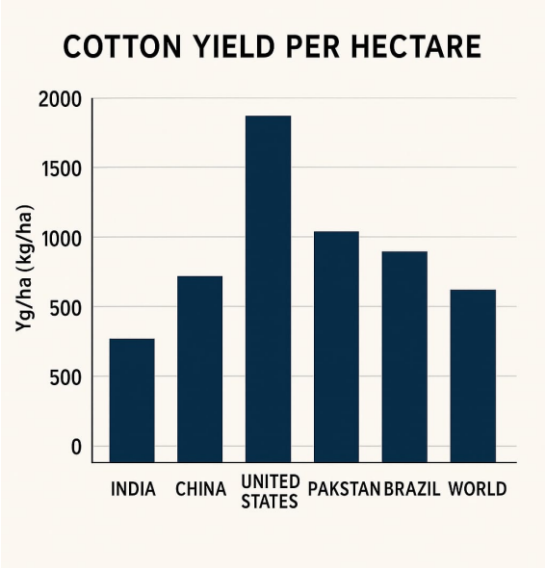
Serviceable Obtainable Market (SOM) – Chennai & Nearby Districts:

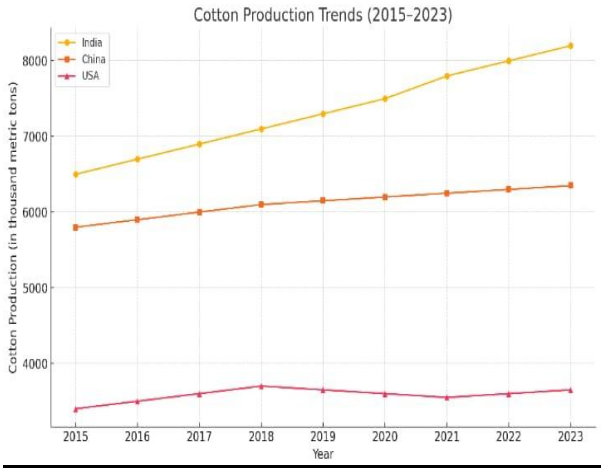
- **Cotton farming area (within 150–200 km from Chennai)**: ~25,000 acres
- **Interested in this solution**: **60%**

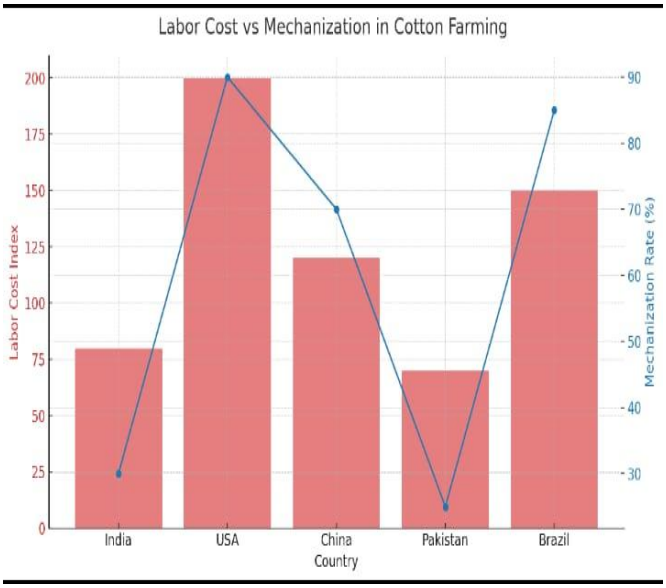
SOM = 25,000 acres × 60% = 15,000 acres
SOM ≈ ₹15 crore

Statistical Data

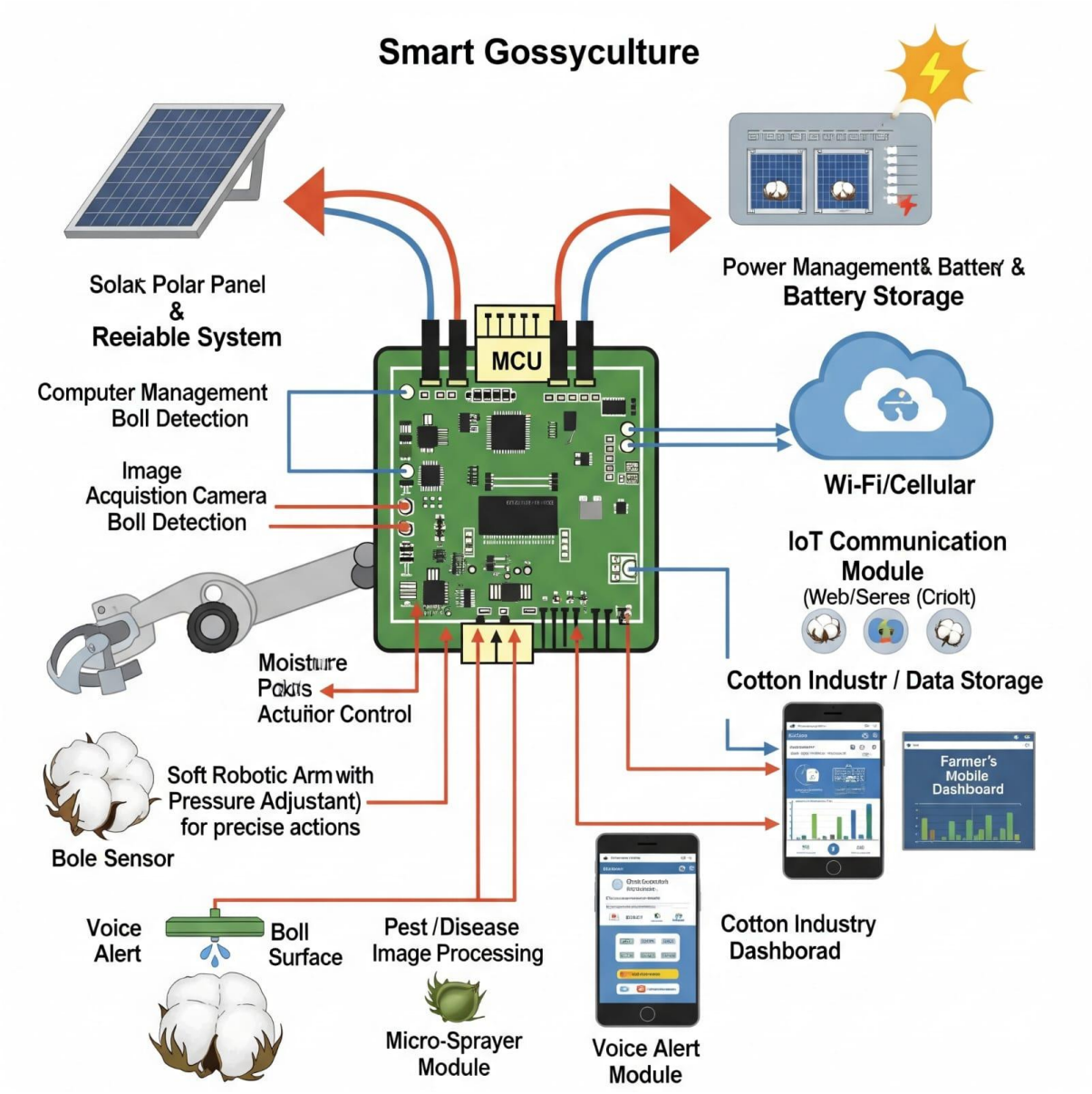








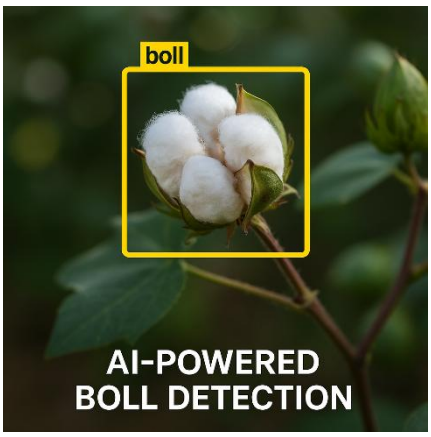
Block Diagram



Expected Outcomes

Proof of Concept (PoC) for the **Smart Gossyculture** project, I'll focus on demonstrating the core functionalities of the system in a simplified version.

1. AI-Based Boll Detection



Goal: Demonstrate the ability of the system to detect cotton bolls using computer vision and machine learning.

Steps:

- **Image Collection**: Gather a dataset of cotton bolls in different stages of growth (using freely available datasets on Roboflow and Kaggle).
- **Pre-processing**: Prepare and annotate the images for training. If necessary, augment the data (rotations, zooms) to improve model accuracy.
- **Model Selection**: Implement **YOLOv5** for real-time object detection. These models are lightweight and efficient, suitable for edge processing.
- **Model Training**: Use a machine learning framework like **TensorFlow** to train the model on the annotated dataset.
- **Integration**: Deploy the model on a **Raspberry Pi** for real-time processing and use a camera module for image capture.

Expected Output: The AI model will recognize mature cotton bolls in real-time and output the position of detected bolls.

2. Moisture-Based Boll Readiness Detection

Goal: Ensure that only **dry and mature cotton bolls** are picked to prevent fiber damage and post-harvest issues.

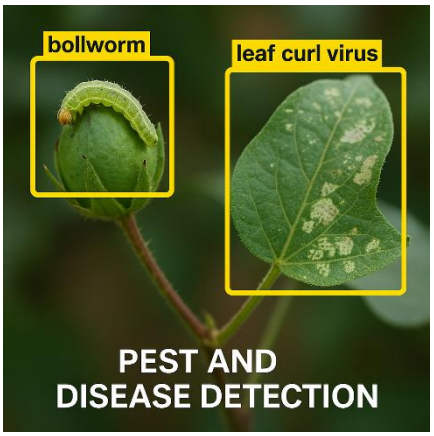
Steps:

- **Sensor Selection:** Use a **resistive surface moisture sensor** [**DHT11/DHT22**] combined with surface contact sensors to detect the moisture level on the boll.
- **Sensor Placement:** Attach the sensor to the robotic arm tip or alongside the camera to check the surface of each detected boll.

- **Threshold Logic:** Set a moisture threshold (e.g., <15%) to determine if the boll is dry enough to be harvested.
- **Integration:** Combine with AI boll detection logic—only send pick command to the robotic arm if the moisture condition is satisfied.

Expected Output: The system will avoid plucking bolls that are too moist (e.g., due to rain or dew), improving cotton quality and reducing post-harvest losses.

3. Pest and Disease Detection



Goal: Implement the system's ability to detect common pests and diseases in cotton plants.

Steps:

- **Dataset Selection**: Utilize available datasets like the **Cotton Pest Dataset** from Roboflow and the **Cotton Disease Dataset** from Kaggle.
- **Model Training**: Similar to boll detection, use **YOLOv8** to train separate models for pest and disease detection.
- **Edge Deployment**: Deploy these models on a single device, a **Raspberry Pi** integrated with a camera to scan for pests and diseases.
- **Notification System**: Once pests or diseases are detected, trigger an alert to the mobile app or a simple on-screen warning.

Expected Output: A detection system that can identify pests like bollworms or diseases like leaf curl virus from images and provide real-time alerts.

4. Soft Robotic Picker Arm



Goal: Demonstrate the prototype of a robotic arm capable of picking cotton bolls.

Steps:

- **Robotic Arm Design**: Build a soft robotic arm using flexible silicone material, which can grip and release cotton bolls without damaging them. A simple gripper mechanism can be created using an actuator and motor for controlled movement.

- **Control System**: Integrate the robotic arm with an **Arduino** to control movements and grip based on boll location (provided by the AI model).
- **Gripper Testing**: Test the gripper on cotton bolls (real or simulated), ensuring that it picks them without damaging the fibers.

Expected Output: A soft robotic gripper that can pick cotton bolls based on real-time location data from the AI system.

5. Automatic Pesticide Sprayer



Goal: To automatically spray pesticide on pest- or disease-affected areas of cotton plants with precision, only when needed, based on AI-based image detection, reducing chemical usage, saving costs, and minimizing environmental and health impacts.

Steps:

- **Detection & Localization:** Using the pest/disease detection AI model (trained via YOLOv5) to identify infected regions in the camera's field of view. The model outputs bounding boxes (coordinates) around infected plant parts.
- **Trigger Decision Logic:** If pests or symptoms are detected with a confidence threshold (e.g., ≥80%), send a signal to the control unit. Use an Arduino to handle the decision logic.
- **Targeting System:** Mount a **micro-sprayer** near the camera or robotic arm. Servo motors adjust the nozzle's angle to align with the detected area (based on image coordinates).
- **Spraying Mechanism:** A small **peristaltic pump** connected to a pesticide reservoir dispenses pesticide through the nozzle. The system sprays only for a short, controlled burst (1–2 seconds) per detected region.
- **Safety and Feedback:** Include a moisture or proximity sensor to avoid spraying in the wrong direction or while moving. Log the event (location, timestamp, image) and notify the farmer via the mobile app.

Expected Output: Precision spraying of affected crop regions with minimal pesticide use. Automated pest control with no manual intervention. Real-time alerts and logs are sent to the mobile app with photos and data on each spray action. Reduction in pesticide cost and health risk for the farmer.

6. IoT Connectivity and Mobile Dashboard



Goal: Demonstrate the ability to collect and display data from the BollBot system on a mobile app.

Steps:

- **Sensor Integration**: Use sensors (e.g., temperature, humidity, soil moisture) connected to the **Arduino** to monitor environmental conditions.
- **Data Uploading**: Set up basic **IoT communication** using **MQTT** to upload sensor data to the local server.
- **Mobile App Development**: Create a basic mobile app using **React Native** that displays real-time data about the boll picking, pest/disease alerts, and environmental conditions.
- **App Integration**: Send data from the edge device (Arduino) to the mobile app, where farmers can monitor system performance and crop health.

Expected Output: A mobile app that displays real-time boll picking performance, pest and disease detection alerts, and environmental conditions.

7. Regional Voice Alerts System

Goal: Enhance accessibility and usability by providing voice alerts in the farmer's regional language (e.g., Tamil, Hindi, Telugu).

Steps:

- **Message List:** Prepare a set of standard voice alerts such as:
  - "பருத்தி பாகை தயாராக உள்ளது – பறிக்கப்படுகிறது" (Tamil: Cotton boll ready – being picked)
  - "பூச்சி கண்டறியப்பட்டது – பூச்சிக்கொல்லி தெளிக்கப்படுகிறது" (Pest detected – spraying pesticide)
  - "பாகை நனையிருக்கிறது – பறிக்க முடியாது" (Boll is wet – cannot pick)
- **TTS Engine:** Use a lightweight Text-to-Speech (TTS) engine **gTTS (Google Text-to-Speech)** for offline/online voice generation.

- **Audio Playback:** Use a speaker module connected to Raspberry Pi/Arduino to play pre-recorded MP3 files or real-time TTS-generated audio.
- **Language Config:** Optionally allow users to choose their preferred language during initial setup via mobile dashboard.

Expected Output: Farmers receive timely voice guidance during boll picking and pest spraying, improving human-machine interaction and reducing training effort**.**

8. Solar-Powered System



Goal: Demonstrate the potential for using solar power to operate the BollBot in off-grid areas.

Steps:

- **Solar Panel Integration**: Connect a small solar panel (sized appropriately for a prototype) to a battery system that can power the Raspberry Pi and other components (camera, motors, sensors).
- **Power Monitoring**: Add a power management system to ensure that the BollBot remains operational in remote locations without reliance on grid power.

Expected Output: A prototype BollBot that runs autonomously on solar power, showing the feasibility of off-grid operations.

9. Final Integration and Testing



Goal: Integrate all components into a cohesive prototype.

Steps:

- **System Assembly**: Combine the boll detection, pest/disease detection, robotic

arm, and IoT components into a working prototype.

- **Testing**: Conduct tests in a controlled environment (e.g., a cotton field or simulated setup) to verify that the system can accurately detect bolls, pests, and diseases, while also picking cotton and sending alerts.
- **User Feedback**: Gather feedback from farmers or experts on the usability and effectiveness of the system.

Expected Output: A fully integrated PoC that can pick cotton bolls, detect pests/diseases, and provide real-time data on a mobile dashboard.

Budget Plan

Category	Sub-Category	Estimated Cost (₹)	Description
I. Hardware & Robotics		**7,00,000**	
	Robotic Arm & Actuators	2,50,000	High-precision, durable soft robotic arm with advanced force-feedback sensors for delicate cotton picking. Includes high-torque servo motors and sophisticated control mechanisms.
	Imaging System	1,80,000	Industrial-grade high-resolution camera (e.g., global shutter, low-light performance) and dedicated edge AI processing unit (e.g., NVIDIA Jetson Xavier NX) for real-time, on-device AI model inference.

	Sensors	70,000	High-accuracy surface moisture sensors (e.g., dielectric sensors) and integrated environmental sensors (temperature, humidity, soil moisture) with industrial-grade reliability.
	Micro-Sprayer System	1,00,000	Precision micro-peristaltic pump, high-accuracy spray nozzle, and advanced targeting mechanism with fine servo control. Includes a durable, chemically resistant reservoir.
	Mobility Platform & Chassis	80,000	Robust, all-terrain robotic chassis with high-torque motors for stable movement across uneven agricultural fields. Includes appropriate suspension and weatherproofing.
	Solar Power System	20,000	High-efficiency monocrystalline solar panel, MPPT charge controller, and a high-capacity deep-cycle battery for extended autonomous operation.

II. Software & AI Development		**4,50,000**	
	AI Model Training & Optimization	2,50,000	High-performance GPU cloud computing for extensive training of YOLOv5/YOLOv8 models for boll, pest, and disease detection. Includes specialized datasets and expert annotation services.
	IoT & Cloud Infrastructure	1,00,000	Premium IoT platform subscription for secure data ingestion, real-time processing, and scalable data storage. Cloud-based backend for mobile application support and analytics.
	Mobile App Development (UI/UX & Features)	70,000	Professional development of a user-friendly mobile application (React Native) with advanced features like detailed dashboards, historical data, customizable alerts, and future integration points.
	Regional Voice Alerts System Integration	30,000	Integration of high-quality, natural-sounding Text-to-Speech (TTS) engine with support for multiple Indian

			regional languages, including robust offline capabilities.
III. Research, Development & Prototyping		**2,50,000**	
	Materials & Fabrication	1,50,000	Costs for rapid prototyping materials (e.g., 3D printing filaments, specialized alloys, custom PCBs), fabrication services, and electronic components for iterative development.
	Testing & Validation	1,00,000	Expenses for conducting rigorous testing in controlled environments and initial field trials to validate system performance, accuracy, and durability.
IV. Contingency		**15,000**	
	Documentation & Presentation Materials	15,000	High-quality visual aids, detailed technical documentation, and professional presentation materials for the hackathon.
Total Estimated Budget		**14,15,000**	

SDG Alignment

- SDG 2: Zero Hunger (increased yield, reduced loss)
- SDG 8: Decent Work & Economic Growth (rural tech employment)
- SDG 12: Responsible Consumption (optimized pesticide use)

Conclusion

This **Proof of Concept** serves to demonstrate the viability of the **Smart Gossyculture** as an AI-integrated solution for automated cotton harvesting with pest and disease detection. While the full-scale implementation will require further refinement, testing, and iteration, the PoC provides a foundational framework to showcase the core functionalities and impact of the system.